A Method to Construct Geographical Crosswalks with an Application to US Counties since 1790*

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Abstract

Empirical researchers often have to map data provided for a "reporting" spatial unit, say, counties in 1900, to a "reference" one, say, counties in 2010. We discuss a general method to create such *crosswalks*: computing the share of the area of each reporting unit nested in a given reference unit. Using these shares, data can be re-aggregated from the reporting to the reference units. We apply the method to construct a crosswalk for mapping US county-level data from 1790 to 2010 to present-day county or commuting zone delineations. We then extend the method by computing the same shares using additional geographic information on livable terrain and historical data on the size and location of towns, which inform the distribution of population within counties in each decade. We also provide the code to generate other crosswalks given maps of reporting and reference units.

Keywords: Regional Analysis, Regional Economic Activity, Regional and Urban History
JEL Codes: A1, C80, N11, R10

*Corresponding author: Fabian Eckert, fpe@ucsd.edu. A repository with the county crosswalks for every decade since 1790, the commuting zone map file and replication code, and general Python code to create crosswalks given any reporting and reference maps can be found at https://www.fpeckert.me/eglp. Users of the materials in the repository are asked to cite this paper as a reference. We are grateful to Henry Glick from the Yale StatLab for help with ArcGIS and to Dave Van Riper from the Minnesota Population Center for numerous discussions about the details of historical geographical delineations in the census data. We also thank David Autor, Rebecca Jackson, Stephen Redding, and Esteban Rossi-Hansberg for early comments and encouragement. Gvirtz thanks the German National Academic Foundation and the Economic and Social Research Council for generous financial support. The present version of this paper supersedes an earlier one with the title "A Consistent County-Level Crosswalk for US Spatial Data since 1790."

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1. Introduction

Many social scientists work with data on regional outcomes, e.g., wages at the county level. However, data sources often report information on different levels of geographical aggregation, making the construction of unified data sets difficult. To map data between different geographies, *crosswalks* are needed, sometimes within a year and sometimes across years. For example, within a year, one data set may report school funding at the school district level, and another may report education outcomes for census tracts. Across years, 1940 population data may only be available for historical county delineations, while 2010 population numbers are available only for present-day delineations. In this paper, we describe a general method to create geographical crosswalks for both scenarios, and implement it in a simple Python routine.

To construct a geographical crosswalk using our method, two maps in the "shapefile" format are required: one for the spatial units for which data are reported (the "reporting" units or map) and one for the spatial units to which the data are to be re-aggregated (the "reference" units or map).[1] Given these two maps, our code calculates a set of "weights" that constitute the crosswalk. Users can then combine these weights with data available for the reporting units, and re-aggregate them to the reference units.

We apply our method to create crosswalks across time for two geographies frequently used in economic research: counties and commuting zones (as delineated by Tolbert and Sizer, 1996).[2] Our crosswalks can be used to map US county-level data from 1790 to 2010 to one of three alternative reference units: 1990 county boundaries, 2010 county boundaries, or 1990 commuting zones. We choose counties to showcase our method since they are the most detailed geographical unit reported in every census since 1790. The crosswalk allows researchers to construct a county-level data set with consistent geographical units for the entire period of existence of the United States. At https://www.fpeckert.me/eglp, we provide code for the general method, replication code for the construction of our county and commuting zone crosswalks, and the crosswalks themselves.

To understand the idea behind our crosswalk, suppose one wants to display data available for a reporting map on the geographical delineations of a reference map. The two maps cover the same territory (e.g., the contiguous United States), but contain different partitions of it (e.g., counties from different time periods or census tracts and states). Our method overlays the two maps using "Geographic Information Systems" (GIS). The intersection of the two maps generates a new map consisting of smaller mutually exclusive spatial "subunits." These subunits can be combined to

[1] Shapefiles are a special data format for maps. Shapefiles contain a collection of polygons oriented in space. Shapefile editors, e.g., ArcGIS or QGIS, can open shapefiles and display the polygons contained in them on the map of the world.

[2] For examples of recent work at this geographical resolution, see Autor and Dorn (2013), Autor et al. (2013), or Acemoglu and Restrepo (2020).

form the original partitions in *either* the reporting or the reference map.

The areas of these subunits can be used to construct weights to re-aggregate data provided for the spatial units of the reporting map to the spatial units of the reference map. The collection of these weights constitutes our crosswalk. Weights are always associated with spatial units in both the reporting and reference map. To map stock data (e.g., population or payroll) the appropriate weights are simply the shares of the subunits in the total area of their spatial unit in the *reporting* map. The underlying assumption is that the variable of interest is distributed proportionally to area size across the subunits contained within each spatial unit of the *reporting* map. To use the crosswalk, one assigns fractions of the variable of interest to the different subunits according to their weights and aggregates these subunits to the spatial units in the reference map.[3]

We apply our method to US county delineations between 1790 and 2010.[4] A key source for historical information on the US economy is the National Historical Geographic Information System (NHGIS) website (see Manson et al., 2019). It provides a host of information (e.g., data on population size, the housing stock, agricultural land, and a variety of other economic variables) for various geographical units from 1790 to the present. However, NHGIS and similar sources report data with geographical identifiers current at the time of data collection. In the case of counties, enormous changes in their number and geographical delineations occurred between 1790 and today. Even counties with the same names may have very different shapes and locations in 1790 and 2010. Geographical changes are mostly due to the territorial expansion of the United States and population growth that led existing counties to split.[5] As a result, data for different decades cannot simply be combined into a decadal panel. For a longitudinal study, researchers require a crosswalk that maps all data to a common set of geographical units. We discuss intricacies specific to the creation of a county crosswalk for the United States and use our method to construct such a crosswalk that maps all counties since 1790 to present-day counties or commuting zones.

The area-based weights assume that the variable of interest, such as population, is distributed

[3]While we intend our crosswalk to be primarily used with data on stocks, weights to map averages (e.g., average wages) can also be constructed. The correct weights, in this case, are found by computing, for each spatial unit in the *reference* map, the share of each of its subunits in its total area. In the crosswalk files provided on our website, we include the weights for stock data. We also provide the areas of all subunits, which can be used to construct weights for data on averages.

[4]The crosswalk described in this paper was constructed as part of the empirical analysis in Eckert and Peters (2022). Eckert and Peters (2022) aggregate historical county-level data to the level of 1990 commuting zones to study the joint movement of US workers across sectors and labor markets since 1880.

[5]Territorial expansion led to the creation of more and more administrative spatial units. Population growth caused existing administrative spatial units to split to maintain a similar population distribution across administrative entities. A host of other factors drove changes in county boundaries over the decades. To borrow an example from Horan and Hargis (1995), the original Williams County, North Dakota, was annexed to Mercer and Stark counties and ceased to exist in 1892. In 1891, in a different section of the state, a "new" Williams County was established. As a result, the same county name appears in two successive decennial censuses (1890 and 1900), but refers to entirely different geographical areas. Horan and Hargis (1995) date the annexation to 1899; we follow the dates in the Atlas of Historical County Boundaries (see Long, 1995).

uniformly within each reporting unit. We extend our approach by relaxing this assumption in two incremental steps. In the first step, we use only the area of each subunit that is not covered by uninhabitable terrain, such as lakes, high elevations, and steep slopes. Specifically, we remove natural water bodies of at least one square kilometer, land above 3,500 meters, and slopes of more than 15 degrees from each subunit. We then compute the same shares using the remaining livable area. We refer to the resulting weights as the livable-area weights.

In the second step, we use historical data on the size and location of towns to estimate the distribution of population within each county in each decade. The data, compiled by Steiner and Heppler (2015), report the towns counted by the census in each decade from 1790 to 2010 and the population of each. We take their locations from the Geographic Names Information System (see U.S. Geological Survey, 2026). In each decade, we assign the population of each town to its location and then distribute the remaining population using information from the subsequent census year. We then compute the same shares using these population estimates. We refer to the resulting weights as the population-based weights.

We compare the area-, livable-area-, and population-based weights with data on where people lived. The comparison uses the Census Place Project (see Berkes et al., 2023), which reports the coordinates of the places recorded in the historical censuses from 1790 to 1940. In the average decade, the area-based weights allocate 5.0% of the population to a different 2010 county than these data, the livable-area weights 4.9%, and the population-based weights 2.7%. The area-based and livable-area weights can nevertheless be useful in applications. They provide a more agnostic benchmark, since they rely on fixed geographic features rather than on historical information about where people lived.

The online data repository associated with this project (https://www.fpeckert.me/eglp) contains Python code to intersect two different maps of the same geography (e.g., a county and a census tract map of the United States) and calculate the collection of weights that constitute a crosswalk between them. Hence, the code is not specific to our application, but can easily be amended to construct crosswalks for arbitrary geographies. In Appendix B, we provide step-by-step instructions for researchers to use our code. The repository also offers our crosswalk for US counties since 1790 and replication files that document its construction. Lastly, it includes example code that applies our crosswalk to map 1900 county-level population data to 2010 county delineations.

We are not the first to construct a crosswalk from historical to present-day counties. Horan and Hargis (1995) produced a "County Longitudinal Template" that allows for an intertemporal comparison of fixed county groups between 1840 and 1990. Rather than trying to map older county delineations to current ones, Horan and Hargis (1995) sought longitudinal comparability by exploiting the fact that many present-day counties are the result of the partitioning of older, larger counties. Their aggregation strategy therefore groups more recent counties that share

common "origins" into larger historical counties that can be compared across time.

Our crosswalk has several advantages over the one provided by Horan and Hargis (1995). First, their approach leads to larger county clusters the longer the time horizon of the analysis. As a result, it does not allow researchers to work with actual present-day counties or commuting zones. For example, to conduct an analysis using data from 1900 to 1990, the data for all decades have to be mapped to the large county groups necessary to accommodate the earliest year, i.e., 1900. In Appendix D, we provide an example of how large these clusters become. Second, Horan and Hargis (1995) rely on information on historical county boundary changes documented by the US Census Bureau in writing. When the boundaries of a county change repeatedly, it is difficult to keep track of where its territory went without a systematic comparison of the respective *maps*. Our method relies instead on the universe of county maps available for the United States to track all changes in county boundaries between census years. We also extend the analysis of Horan and Hargis (1995) 50 years backward (from 1840 to 1790) and 20 years forward (from 1990 to 2010). The clusters of Horan and Hargis (1995), in turn, need no assumption about how a variable is distributed within a county, while our crosswalks assume that it is distributed in proportion to area, to livable area, or to estimated population (see Sections 2 and 3.3).

The structure of this paper is as follows. Section 2 describes and discusses the method to construct crosswalks. Section 3 applies our method to construct a crosswalk from US counties since 1790 to present-day geographies. We also provide a brief overview of the territorial expansion of the United States, the evolution of its county delineations, and the resulting challenges in constructing a crosswalk for counties. That section also introduces the livable-area and population-based weights. Section 4 maps historical county populations to present-day counties and shows how much the choice of weights matters. Section 5 concludes.

2. A Simple Method to Construct Geographical Crosswalks

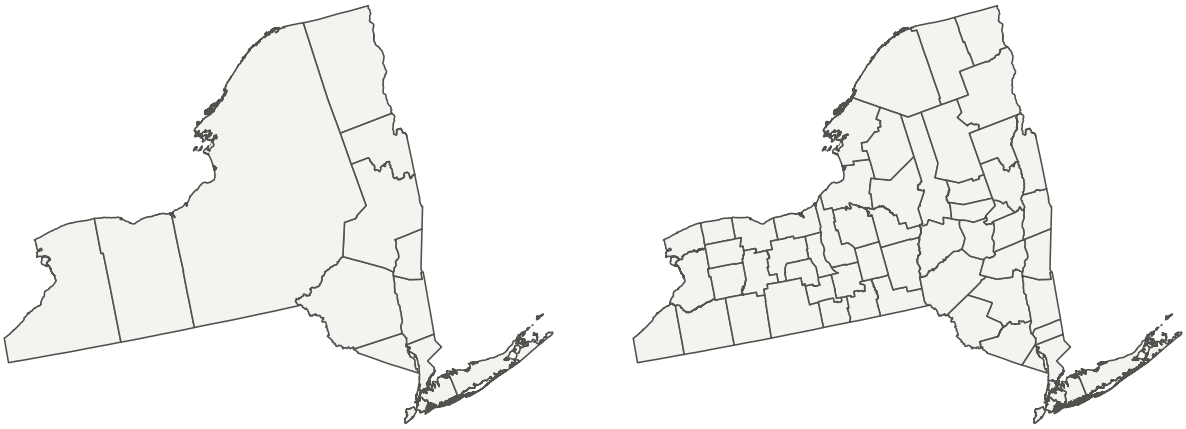
In this section, we describe our method to construct geographical crosswalks.

2.1 The Method: Intersecting Different Maps of the Same Territory

Our crosswalk is the result of a simple accounting exercise. We intersect two different maps covering the same territory but partitioned in different ways. As an example, consider the two maps of New York State in Figure 1. The figure displays the county maps for New York in 1790 (left panel) and in 2010 (right panel). Figure 2 shows the intersection of these two maps. The thick black lines show the 1790 county boundaries, and the thin dashed blue lines show the 2010 boundaries.

The subunits resulting from the intersection cover the entire area of New York State and are

Figure 1: New York State County Boundaries in 1790 and 2010



(a) 1790 Counties (b) 2010 Counties

Notes: The figure shows the land area of New York State. The left panel shows the county boundaries in New York State in 1790. The right panel shows the county boundaries in New York State in 2010. The underlying shapefiles are taken from Manson et al. (2019).

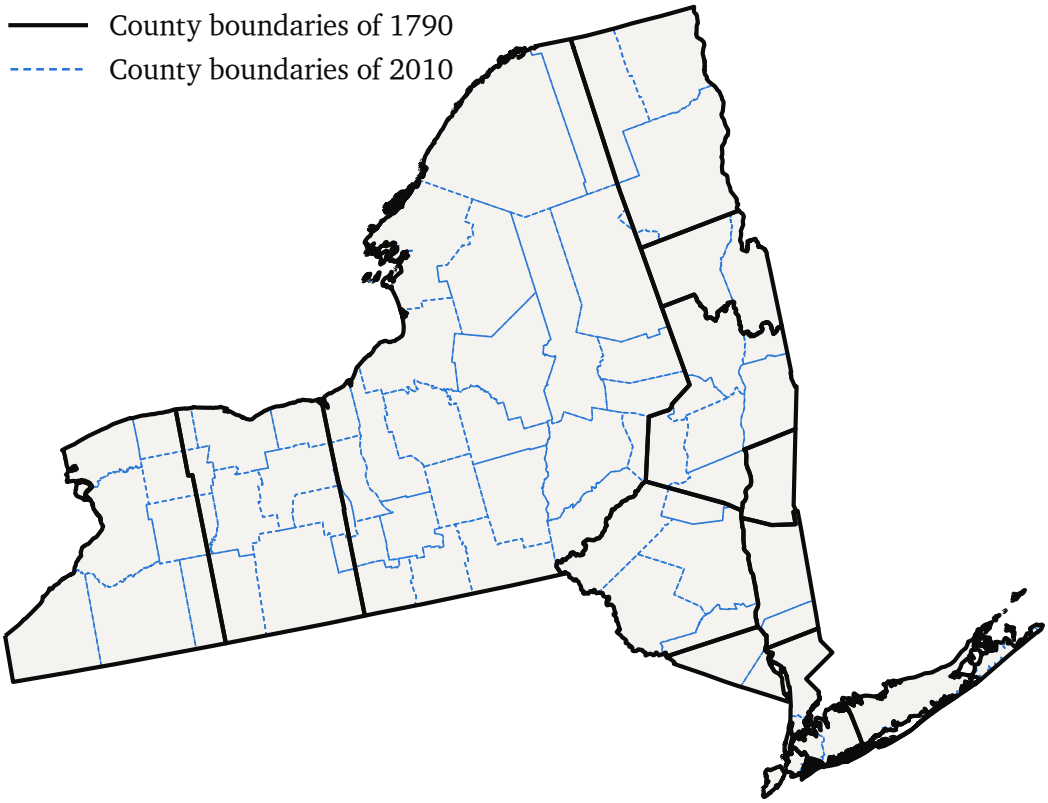
pairwise disjoint. They can therefore be combined to form any spatial unit in either of the original maps. We use these subunits to construct a collection of weights that constitutes our crosswalk by considering each spatial unit in the *reporting* map and computing the share each of its subunits has in its total area. Each weight is associated with a spatial unit in both the reporting and the reference map and each spatial unit in the two maps has at least one weight associated with it.

To see how to use the resulting crosswalk, consider a specific variable, say population counts by county in 1790. By multiplying the 1790 county population counts for a given county by the associated weights, we distribute its population across the subunits it contains in proportion to their area. We can then sum these population counts across the subunits that are contained within each of the 2010 county delineations. The result is a set of 1790 population counts at the level of 2010 county delineations.

The underlying assumption is that the variable of interest is distributed proportionally to area size across the subunits contained within each spatial unit of the *reporting* map. The plausibility of the assumption depends on the variable of interest. To achieve maximum accuracy, researchers should always use the most disaggregated maps for which data are reported when constructing geographical crosswalks.

The method does not require the use of area, however. The same subunits can be weighted by any measure available for them, such as their livable area or an estimate of their population. Section

Figure 2: Intersection between New York State County Boundaries in 1790 and 2010



Notes: The figure shows the land area of New York State. The thick black lines show the county boundaries of 1790. The thin dashed blue lines show the county boundaries of 2010. The figure shows the spatial subdivisions formed by intersecting the 1790 county map with the 2010 county map. The underlying shapefiles are taken from Manson et al. (2019).

3.3 applies this approach to US counties.

We recommend using our method primarily with stock data (e.g., data on population or total payroll at the regional level). To map data on average outcomes (e.g., average wages) between different spatial units, a different set of weights is required. They are obtained by considering each spatial unit in the *reference* map and computing the share each of its subunits has in its total area.[6]

The Python code we provide only produces weights for stock variables since their use is subject to fewer assumptions. The county crosswalk files we provide do, however, report the area of each

[6]In this case we need to assume that the weighting variable underlying the average (e.g., workers in the case of average wages) is distributed proportionally to area size across the subunits contained within each spatial unit of the *reference* map *and* that the averages are constant across the spatial units in the *reporting* map. Note that when the weighting variable is observed, it can always be used to construct stocks (e.g., total payroll by combining total workers with average wages) and our crosswalk can be used as is. We recommend this approach.

Table 1: Examples of a County Crosswalk in 1800 and 1810

Panel A: The Crosswalk for 1810

	1810		2010		
Year	State	County	State	County	Weight
1810	1	A	1	A	1
1810	1	B	1	B	1
1810	1	C	1	C	1

Panel B: The Crosswalk for 1800

	1800		2010		
Year	State	County	State	County	Weight
1800	1	A	1	A	1
1800	1	B	1	A	0.1
1800	1	B	1	B	0.9
1800	1	C	1	C	1

Notes: The table contains an example of the construction of our crosswalk. Panel A contains a hypothetical example of 1810, where the historical counties map directly into the current county borders. Panel B contains an example of 1800, where the original county B was split between the current counties A and B.

subunit in the intersected map, which researchers can use to compute weights for data on average outcomes. For the rest of the paper, all references to "crosswalk" apply to the collection of weights needed to aggregate stock variables.

Appendix A discusses more details of the crosswalk construction and its use. Appendix B describes how to use the Python implementation of our method to construct a crosswalk given any reporting and reference maps in the shapefile format.

2.2 Using the Crosswalk

To illustrate our crosswalk, consider the following hypothetical example. Take a map of counties in 1800, one in 1810, and one in 2010. The goal is to create a crosswalk mapping data provided for the 1800 and 1810 counties to the 2010 county delineations.

Suppose county A took over 10% of county B's land in 1810 and that, since then, there were no further changes. In a first step, we intersect the 2010 county map with the 1810 county map. We notice that the county boundaries overlap perfectly and assign a weight of 1 to each county in

1810, implying the data in 1810 need not be adjusted. See Panel A of Table 1 for the resulting crosswalk for these counties.

Next, we intersect the 2010 map with the 1800 map. We notice that the 2010 county A consists of the 1800 county A and 10% of the 1800 county B. To account for this, we add a second line for county B as shown in Panel B of Table 1. The first line with a weight of 0.1 indicates the share of the 1800 county B mapping to the 2010 county A; the second line with a weight of 0.9 indicates the share of the 1800 county B that maps to county B in 2010. To use the crosswalk, researchers should allocate 10% of the population recorded for county B in 1800 to county A in 2010, and 90% to county B.

2.3 Implementing the Method in Python

We implement our method in Python using the "GeoPandas" package.[7] Our program calculates the weights that constitute the crosswalk for two arbitrary shapefiles of the same geography with different partitions (e.g., counties and states of the US). Each of the two maps needs a "coordinate system," a property of GIS shapefiles; the code projects both maps to a common projection that preserves areas. Users only need to specify which map is the reporting and which the reference map. The code then computes the weights necessary to aggregate data available for the reporting units to the reference units. The online repository associated with this paper (https://www.fpeckert.me/eglp) contains the Python routine.

3. Application: Mapping Historical US County Data to the Present

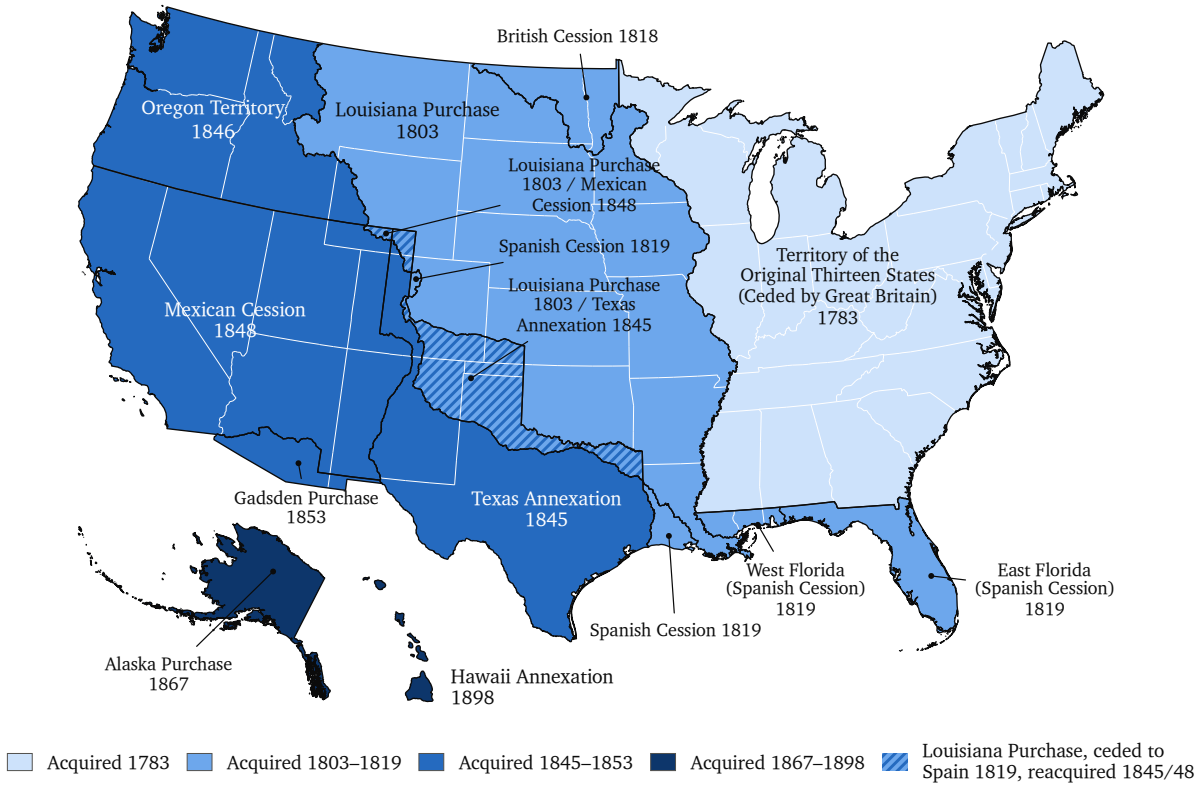
In this section, we apply our method to construct a crosswalk for US counties since 1790. We first provide a short overview of the history of the US territory and its counties. Then we offer details on the crosswalk construction and discuss caveats. Lastly, we introduce crosswalks based on livable area and on estimated population.

3.1 The Evolution of the Map of US Counties

Three features of US history are responsible for the vast changes in the geographical delineations of its counties. First, the territorial expansion of the US throughout the nineteenth century increased its territory significantly and led to the creation of *new* counties. Second, the conversion of existing US "territories" to "states" was usually associated with more detailed spatial administrative units. Third, the existing county borders were adjusted as the population grew. We discuss them in turn.

[7] GeoPandas is an open-source project that facilitates work with geospatial data in Python. For the GeoPandas documentation, see http://geopandas.org/.

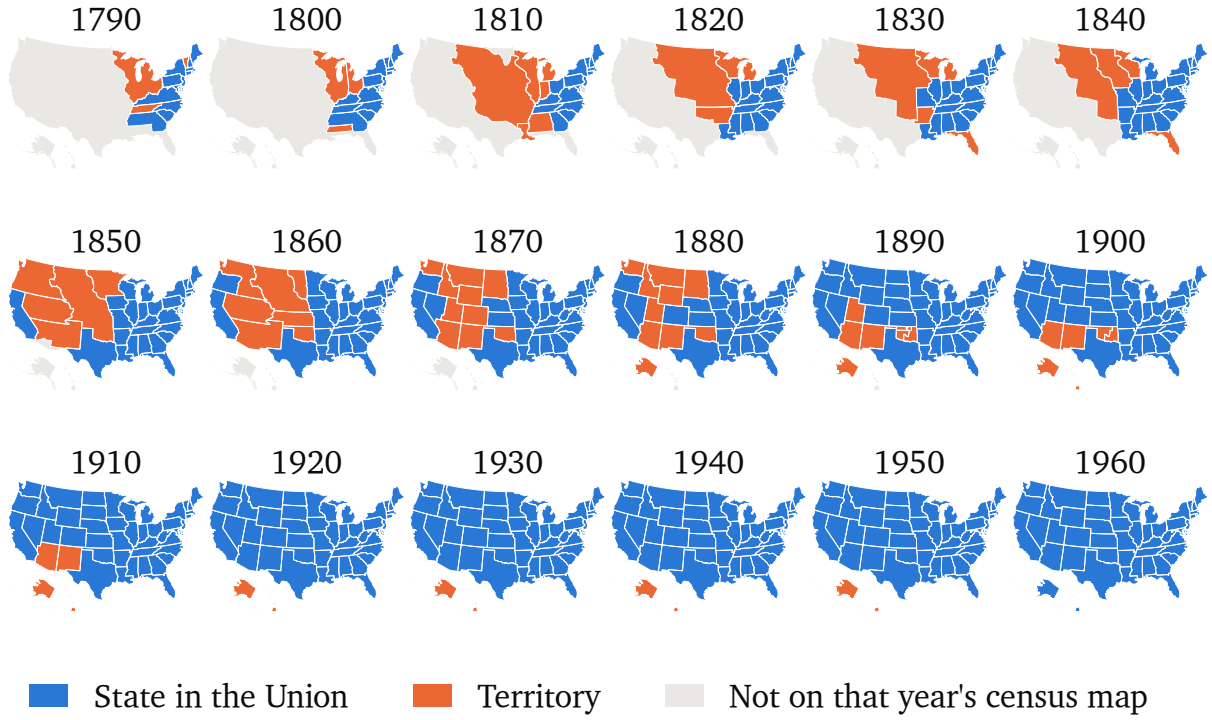
Figure 3: Territorial History of the United States



Notes: The map shows the areas of today's United States by the acquisition through which they became part of it. The names, dates, and boundaries are those of the map layer "Territorial Acquisitions of the United States" in National Atlas of the United States (2011). The hatched areas were part of the Louisiana Purchase, were ceded to Spain in the Adams–Onís Treaty of 1819, and were reacquired with Texas in 1845 or with the Mexican Cession in 1848. Alaska and Hawaii are shown as insets, Alaska at about a third of its size; Puerto Rico and the Virgin Islands are not shown. The state borders are taken from Manson et al. (2019).

Territorial Expansion In 1776, the thirteen British colonies on the North American continent declared their independence, and the Treaty of Paris of 1783 set the initial territory of the United States. Between then and now the *territory* of the United States grew gradually into its present shape. Figure 3 depicts the major historical milestones of US territorial history, with the names the National Atlas of the United States gives them. The Louisiana Purchase from France in 1803 constituted the first major territorial expansion. The British Cession of 1818 and the Spanish Cession of 1819, which included East and West Florida, followed. Five more acquisitions between 1845 and 1867 gave rise to the US territory as we know it: the Texas Annexation of 1845; the Oregon Territory, which the United States and Great Britain had occupied jointly and divided in the Oregon Treaty of 1846; the Mexican Cession of 1848 after the Mexican–American War; the Gadsden Purchase from Mexico in 1853; and the Alaska Purchase from Russia in 1867. Lastly, Hawaii was annexed in 1898.

FIGURE 4: THE TERRITORY AND THE STATES OF THE UNION OVER TIME



Notes: The figure shows, for each census year from 1790 to 1960, the areas in states that were part of the Union, the areas in territories, and the areas not on that year's census map. Most of the latter were not yet part of the United States; some were, such as Alaska, bought in 1867, which first appears on the census map of 1880. A state counts as part of the Union from the first census in or after its year of accession (see Table E.2). Vermont, an independent republic until it joined the Union in 1791, is shown with the territories in 1790. Alaska and Hawaii are shown as insets. The underlying shapefiles are taken from Manson et al. (2019).

The Creation of the Union Large parts of the United States, however, remained so-called "territories of the United States" even after becoming part of the US proper; that is, they were not yet part of the Union.[8] Figure 4 visualizes the acquisition of new territory and the gradual conversion of these territories into organized states. Every decade before 1920, and again in the 1950s, saw new states or changes in the territorial extent of states. The contiguous United States reached its present extent with the Gadsden Purchase of 1853, and the *territory* as we know it today, including Alaska and Hawaii, was complete by 1898; the census maps cover all of it from 1900 on. However, the *Union* of states reached completion much later with the joining of Hawaii in 1959. Table E.2 in Appendix E provides a list of the dates of accession to the Union for each state.

[8]In fact, 31 of the United States' current 50 states were at one time part of such territories. The six states that were not part of the founding states and were never territories are: Kentucky and West Virginia (both split off from Virginia); Maine (split off from Massachusetts); California (created as a state out of the unorganized territory of the Mexican Cession); and Vermont and Texas (both previously self-declared republics).

Changing County Boundaries In response to changing social, economic, and political conditions within the US, counties have been created, eliminated, and changed their shape over its history. The partitioning of existing counties to form new ones is the most frequent source of changes in geographical delineations.[9]

Table 2 shows statistics on how the number and shape of counties in the United States have changed throughout its history. The second column lists the number of historical subdivisions in the county shapefiles obtained from Manson et al. (2019). These subdivisions are mostly counties, but also include parishes and independent cities that covered part of the US territory in lieu of counties in various decades. Between 1790 and 2010 the number of such units increased by a factor of more than 10, from 292 to 3,143. Their increase is both due to the territorial enlargement of the US and the splitting of existing counties. Column 3 of the table shows the fraction of today's US territory that the county map of each decade covers, territories included. In 1790 the map covered only 22% of today's territory; the census maps cover the full territory only from 1900 on. Columns 4, 5, and 6 provide a measure for the amount and size of changes in county boundaries in each decade. If two maps perfectly overlap, all weights are 1. If there are primarily small changes, many weights will be very small (0.01, say) and many others very large (0.99, say). In a year with large changes, there will be more weights in the intermediate range, e.g., between 0.2 and 0.8. Columns 4, 5, and 6 show the fraction of weights that lie in such intermediate ranges.

3.2 Constructing a County Crosswalk

Our county crosswalk is based on publicly available digital maps of US counties (i.e., shapefiles) for each decade from 1790 to 2010. We downloaded these shapefiles from the NHGIS website (see Manson et al., 2019).[10] For 2010, we use the 2010 "Topologically Integrated Geographic Encoding and Referencing" (TIGER)/Line shapefile, created by the Census Bureau. For all other years, we use the 2000 TIGER/Line shapefiles.[11] For each decade, the shapefiles cover all states in the Union; in addition, they often contain counties in US territories outside of all states in the Union in that year.[12]

We apply the methodology outlined in Section 2 in the following way: for each decade from 1790

[9] See the Atlas of Historical County Boundaries described in Long (1995) for a detailed description of all county boundary changes. The online version of the Atlas can be found at http://publications.newberry.org/ahcbp/index.html.

[10] The NHGIS data repository is updated every year and in future versions some details of the shapefiles may change. However, NHGIS assigns a Digital Object Identifier (DOI) code to each version of its data and archives it. The version history for NHGIS is found at https://ipums.org/projects/ipums-nhgis. We used the 2019 version of the shapefiles, which has been assigned the following DOI code: 10.18128/D050.V14.0. The county population data in Section 4.1 and Appendix E come from Version 21.0 (Schroeder et al., 2026), downloaded in November 2026.

[11] NHGIS provides the 2010 county map only on the 2010 TIGER/Line shapefiles. For more details on the TIGER system, see Marx (1986).

[12] The counties in these territories tend to be large and often change significantly upon accession to the Union. Arkansas, which joined the Union in 1836, serves as a fitting example. In the 1840 shapefile, Arkansas contains 39 counties. A decade earlier, in 1830, the Arkansas "territory" contained only 23 counties.

Table 2: Changing County Numbers and Boundaries

Year	Total Subdivisions	Fraction of Today's Territory	Fraction of Crosswalk Weights *between* 0.4 and 0.6	0.2 and 0.8	0.01 and 0.99
1790	292	0.222	0.018	0.065	0.265
1800	427	0.232	0.021	0.085	0.291
1810	587	0.488	0.020	0.071	0.278
1820	779	0.480	0.021	0.075	0.306
1830	1,002	0.495	0.022	0.084	0.259
1840	1,285	0.495	0.021	0.078	0.210
1850	1,632	0.830	0.015	0.059	0.195
1860	2,126	0.838	0.012	0.042	0.157
1870	2,334	0.838	0.010	0.036	0.129
1880	2,614	0.998	0.010	0.028	0.085
1890	2,799	0.998	0.007	0.020	0.062
1900	2,848	1.000	0.006	0.018	0.050
1910	2,963	1.000	0.003	0.012	0.035
1920	3,075	1.000	0.001	0.003	0.017
1930	3,110	1.000	0.000	0.001	0.009
1940	3,108	1.000	0.000	0.001	0.008
1950	3,111	1.000	0.000	0.001	0.007
1960	3,134	1.000	0.001	0.001	0.007
1970	3,141	1.000	0.001	0.001	0.005
1980	3,137	1.000	0.000	0.001	0.003
1990	3,141	1.000	0.000	0.000	0.002
2000	3,141	1.000	0.000	0.000	0.002
2010	3,143	1.000	0.000	0.000	0.000

Notes: The table shows statistics describing the territorial expansion of the United States throughout its history and the changes in county delineations. The "total subdivisions" count gives the number of counties and other subdivisions (parishes, independent cities) on a given decade's county map. We report spatial units available on the entire territory of the United States for which data are available, not just subdivisions of states that are part of the Union in a given year. Column 3 gives the fraction of today's US territory covered by a given decade's county map, measured as the area of that decade's county map relative to the map of 2000, capped at 1 (the map of 2010 is drawn from different line work and is 0.2% larger). We include territories that were not yet states in the area calculation. Columns 4, 5, and 6 give the fraction of all weights in the county crosswalk that maps a decade's county delineations to the 2010 ones that lie strictly within the stated range. For two identical maps all weights are 1. The larger a fraction in the last three columns, the more the county delineations of that year differ from the county map of 2010.

to 2000, we intersect the corresponding county shapefile with the 2010 county shapefile. For each county in each decade, we then compute the shares of its area that map to the various counties in the 2010 shapefile. Where a small part of a historical county lies outside the 2010 map, for example because the two maps draw a coastline differently, the shares of the part inside are scaled up to sum to one.[13] For each historical county, our crosswalk contains these shares, or weights, and lists the corresponding 2010 county.[14] Since many historical counties map into several 2010

[13]These areas include inland lakes and reservoirs, but not the ocean, bays, or the Great Lakes; see Appendix A.

[14]By present-day counties we mean the counties of the 2010 census. A few counties have changed since then.

counties, the crosswalk often contains many more entries than there are historical or present-day counties. In addition, we repeat the exercise using the maps of 1990 counties and of 1990 commuting zones (see Tolbert and Sizer, 1996) as reference maps.[15] The collection of weights for each decade and each reference map (1990 counties, 2010 counties, 1990 commuting zones) is available in the online repository associated with this paper (https://www.fpeckert.me/eglp).

Table 3 shows the number of 1990 counties that can be mapped to the respective territory of the United States in any given year. We count a county only if at least 1% of its area lies in that territory. Without this rule, the counts would include counties that touch the territory only through slivers, the thin pieces created where the line work of two maps differs slightly (see Appendix A). Since the reference map is the same in every decade, the column of 1990 counties traces the territorial expansion of the United States as the census maps record it. The number of 1990 counties increases the most between 1800 and 1810, after the Louisiana Purchase, and between 1840 and 1850, after the annexation of Texas, the Oregon Treaty, and the Mexican Cession (cf. Table 2 above). The count falls between 1810 and 1820 because the Adams–Onís Treaty of 1819 gave Spain the western parts of the Louisiana Purchase, in today's Colorado, Kansas, New Mexico, Oklahoma, Texas, and Wyoming; they return with the annexation of Texas and the Mexican Cession. The same change explains the dip in column 3 of Table 2. Once the census maps cover the full territory of the US in 1900, the count no longer changes. The other two columns repeat the exercise for 2010 county delineations and 1990 commuting zones, respectively.

Appendix C provides an overview of the variables in the crosswalk files we provide online and how to use them. Appendix D compares our county crosswalk to the one by Horan and Hargis (1995).

3.3 Extensions: Livable Area and Population

We extend our county crosswalk by replacing the area of each subunit with two alternative measures: its livable area and an estimate of its population. The crosswalk so far rests on the assumption that the variable of interest is distributed proportionally to area size across the subunits contained within each spatial unit of the reporting map. This assumption provides a transparent benchmark. However, it allocates part of a county's population to lakes and mountain ranges, and it does not account for the concentration of population in towns. Both extensions use the same reporting and reference maps as before and differ in how the weights are computed.

Bedford City (Virginia) became part of Bedford County in 2013. Petersburg Borough (Alaska) was created in 2013 from part of Petersburg Census Area and part of Hoonah-Angoon Census Area; the rest of Petersburg Census Area went to Prince of Wales-Hyder Census Area. In 2015, Shannon County (South Dakota) was renamed Oglala Lakota County and Wade Hampton Census Area (Alaska) was renamed Kusilvak Census Area, both with new FIPS codes. In 2019, Valdez-Cordova Census Area (Alaska) was split into Chugach Census Area and Copper River Census Area. In 2022, Connecticut replaced its eight counties with nine planning regions as county equivalents. Researchers who combine the crosswalk with data for later years need to account for these changes.

[15] We use the 1990 commuting zone shapefile provided by the Health Inequality Project at https://healthinequality.org/data/.

Table 3: Counties and Commuting Zones throughout US History

Year	1990 Counties	2010 Counties	1990 Commuting Zones	Year	1990 Counties	2010 Counties	1990 Commuting Zones
1790	1,513	1,511	295	1910	3,141	3,143	741
1800	1,558	1,556	305	1920	3,141	3,143	741
1810	2,513	2,511	567	1930	3,141	3,143	741
1820	2,475	2,473	546	1940	3,141	3,143	741
1830	2,542	2,540	561	1950	3,141	3,143	741
1840	2,535	2,533	560	1960	3,141	3,143	741
1850	3,109	3,107	722	1970	3,141	3,143	741
1860	3,111	3,109	722	1980	3,141	3,143	741
1870	3,111	3,109	722	1990	3,141	3,143	741
1880	3,136	3,138	737	2000	3,141	3,143	741
1890	3,136	3,138	737	2010	3,141	3,143	741
1900	3,141	3,143	741				

Notes: The table shows the number of contemporary spatial units our crosswalk maps to the territory of the United States in various decades, including territories that were not yet states. The table hence provides an overview of how many spatial units can consistently be studied from each decade to the present. The table is printed in two halves, side by side. The first column of numbers in each half shows the number of 1990 county delineations that can be mapped to the US territory in each decade. The other two columns of each half do the same for 2010 counties and 1990 commuting zones (see Tolbert and Sizer, 1996). A spatial unit is counted if at least 1% of its area lies in the territory covered by the county map of that decade. This rule excludes units that touch the territory only through slivers created where the line work of two maps differs slightly.

For both extensions, we subdivide the map of the contiguous United States into *grid cells*, squares of 500 by 500 meters. There are about 31 million grid cells, and they are the same in every decade. In each decade, we assign each grid cell to the county in which its center lies, both on the map of that decade and on the reference map. Each grid cell therefore belongs to exactly one subunit. The grid cells are useful because every census map draws different county boundaries. They give all maps a common set of small areas in which to record land and population and to carry estimates from one census to the previous one. Appendix F provides details on the crosswalk construction.

Livable Area The livable-area weights use data on terrain and water. We take elevation and slope from Danielson and Gesch (2011). We take the locations of water bodies and major dams from the National Atlas (see National Atlas of the United States, 2006, 2014).

We classify land as not livable if it lies in a natural water body of at least one square kilometer, above 3,500 meters, or on a slope of more than 15 degrees. We use the dams to tell reservoirs apart from natural lakes. We treat reservoirs as land, since the land they cover was dry before the dams were built. We record the share of the land of each grid cell that is livable. The livable

area of a subunit is the livable land of its grid cells. We use the same map of livable land for all decades. The weight of each subunit is its share in the total livable area of its spatial unit in the reporting map.

Population The population-based weights use data on towns and on county populations that are available for each decade from 1790 to 2010. We take the population of each town in each decade from Steiner and Heppler (2015) and its location from the Geographic Names Information System (see U.S. Geological Survey, 2026).[16] Steiner and Heppler (2015) compile the towns mainly from a Census Bureau list of places that reached 2,500 people, so smaller places count as rural. We take the population of each county from the census (see Schroeder et al., 2026). We define the *rural population* of a county as its total population minus the population in towns in that county and year provided by Steiner and Heppler (2015).

We estimate the population of each grid cell in a given year in two steps. First, in each decade, we place the population of each town in the grid cell at its location. Second, we distribute the rural population of each county across the grid cells that lie in the county.

We construct the estimates of the population of each grid cell backward in time, starting from the most recent census. In 2010, we distribute the rural population of each county uniformly over its livable land, since there is no later census to draw on.[17] In 2000, we distribute the rural population of each county in proportion to the rural population that our 2010 estimate assigns to the same grid cells. In 1990, we use the 2000 estimate in the same way, and so on, one census at a time, back to 1790. In every decade, each county keeps its own census population. The following census only determines how this population is distributed within the county. This procedure extends the logic of the target-density weighting of Schroeder (2007), which uses the distribution of population in one census to allocate the population of another, to many decades.

Because we construct the estimates backward in time, the estimate for each decade carries information from all later censuses. When a county is divided, the rural populations of the new counties determine how the rural population of the earlier county is split between their areas. This split carries over to every earlier decade in which the land formed part of a single county. Later divisions refine the split within each of these areas in the same way. As a result, the distribution of a county's rural population in 1790 draws on every later census that divided its land more finely. Only land that stays within a single county until 2010 receives rural population in proportion to its livable area.

The estimated population of a subunit is the population of its grid cells. The weight of each subunit

[16] Where a town has no match there, we use the coordinates in Steiner and Heppler (2015).

[17] For 2010, more granular information about the distribution of people across space is available. Since such information is not available for most earlier years, we decided against using this information. Instead, every decade between 1790 and 2010 draws on the same data sources and information.

is its share in the total estimated population of its spatial unit in the reporting map.

Figure 5 illustrates the construction of the population-based weights with a made-up example in three steps. Each step uses more information than the one before. Suppose that a county of 1790, A, has 10,000 people and overlaps three counties of 2010, X, Y, and Z, which cover 20%, 40%, and 40% of its area. The figure divides the county into its grid cells, drawn as small squares. The area-based weights spread these people evenly over the county, so X receives 2,000 people, Y 4,000, and Z 4,000 (step 1).

Step 2 of the example adds a town. Suppose that 5,000 of the 10,000 people of county A live in a town in X. The population-based weights first place these 5,000 people in X. Spreading the remaining 5,000 rural people in proportion to area would halve the total population of Y and Z, to 2,000 each, and increase that of X to 6,000 people (step 2).

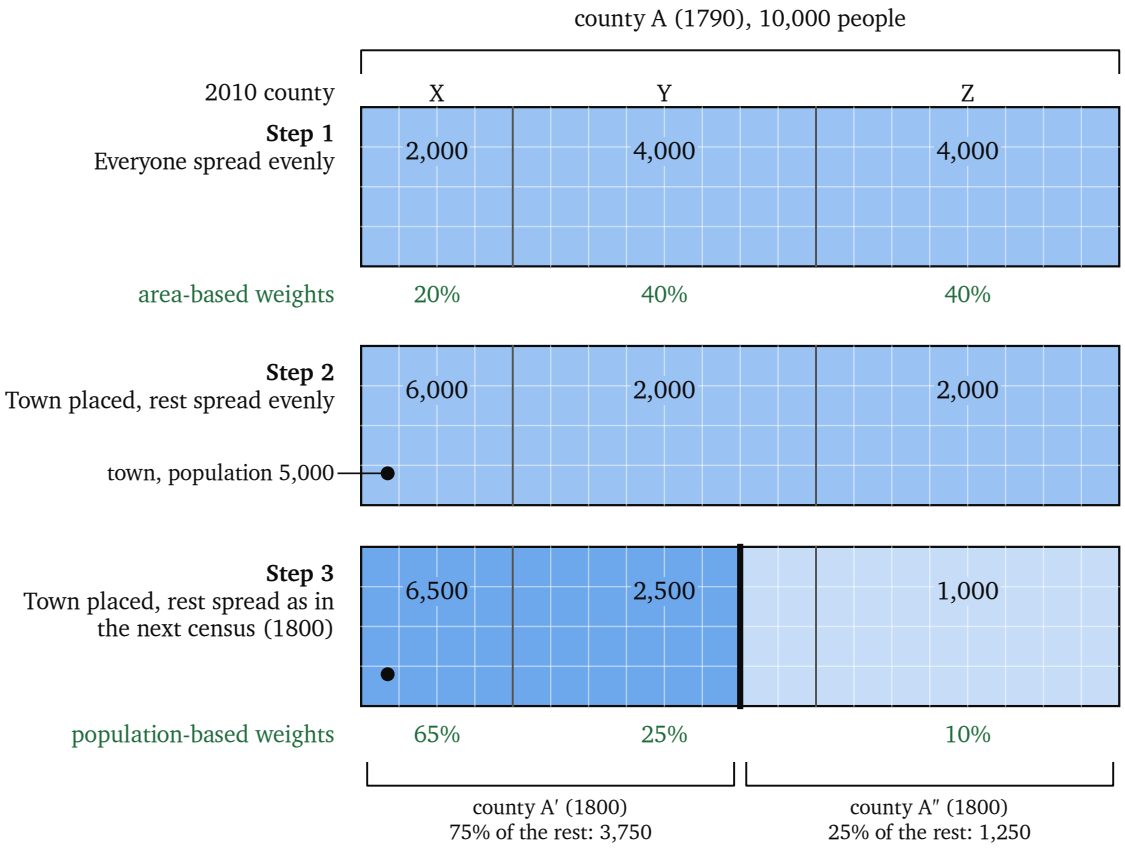
Step 3 of the example spreads the rural people by the next census, of 1800, instead of by area. We assume that by 1800, county A has split into two counties: A′, which covers X and three quarters of Y, and A′′, which covers the remaining quarter of Y and Z. We also assume that neither A′ nor A′′ is divided later. Our estimate of the rural population of 1800 is then even within each of them, since no later census separates one part of A′, or of A′′, from another. Suppose instead that A′ had been divided in 1810. Our estimate for 1800 would then distribute the rural population of A′ unevenly across its grid cells, following the 1810 estimate. Step 3 would use that uneven estimate.

In step 3, the rural population of 1790 is spread over the grid cells of the county in proportion to our estimate of the rural population of 1800. We assume that in the 1800 census, 75% of the rural population of the two counties lives in A′ and 25% in A′′. We therefore assign 3,750 of the 1790 rural people to the grid cells of A′ and 1,250 to the grid cells of A′′. Since the estimate for 1800 is even within each county, so is this assignment.

After step 3, X receives 6,500 of the 10,000 people of county A (the town's 5,000 and 1,500 rural people), Y 2,500, and Z 1,000. The population-based weights are the shares of the county's population: 65% for X, 25% for Y, and 10% for Z. Researchers apply them to any variable observed for county A in 1790, such as its payroll, in the same way as the area-based weights.

The livable-area and population-based weights cover only the contiguous United States, since our map of livable land does not cover Alaska and Hawaii. The area-based crosswalks also cover Alaska from 1880 and Hawaii from 1900. Appendix F discusses more details of their construction and our corrections to the underlying data.

FIGURE 5: CONSTRUCTING THE POPULATION-BASED WEIGHTS

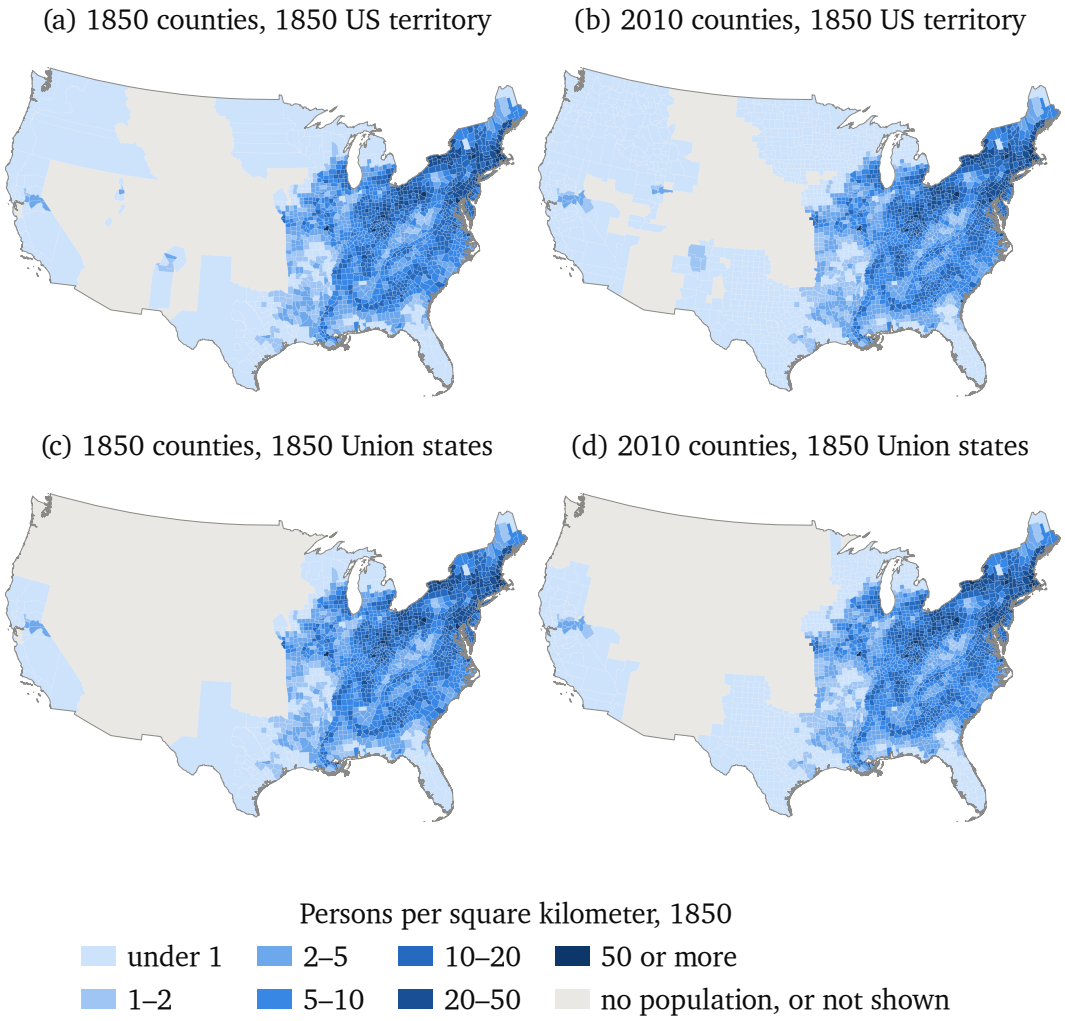


Notes: Each step shows the same made-up county of 1790 (rectangle), its grid cells (small squares), and the three counties of 2010 it overlaps (X, Y, and Z; thin lines). We assume that all of its land is livable. The figure labels the rural population, the people outside the town, as "the rest." The shading shows the share of the people outside the town (in step 1, of all people) that each grid cell receives, darker where the share is larger. In steps 1 and 2, every grid cell receives the same share; in step 3, grid cells in A′ receive three times the share of grid cells in A″. The numbers give the people that each set of estimates assigns to X, Y, and Z; in X, they include the 5,000 people of the town (dot). Below steps 1 and 3 are the area-based and the population-based weights: the numbers in the step divided by the 10,000 people of county A. Each column compares one county of 2010. In step 3, the thick line is the boundary between the counties A′ and A″ into which we assume county A has split by 1800; the bracket below gives their shares of the rest in 1800 and the number of the 1790 rest assigned to each. The county and all numbers are made up.

4. Mapping Historical US County Populations to Present-Day Counties

In this section, we use our crosswalks to map historical US county populations to present-day counties. We first construct a longitudinal panel with the area-based crosswalk and then discuss how much the choice of weights matters.

FIGURE 6: POPULATION DENSITY IN 1850
FOR DIFFERENT COUNTY DELINEATIONS



Notes: The figure shows the population of the United States in 1850 on four maps, as persons per square kilometer in the same classes in all panels. Panel (a) shows all counties and other units on the 1850 map at their 1850 delineations; panel (b) shows the same population mapped to 2010 counties with our area-based crosswalk. Panels (c) and (d) repeat this for the states that were part of the Union in 1850. For 2010 counties, the density is computed over the part of the county covered by the 1850 units shown. Gray areas lie outside the panel's area or report no population; in panels (a) and (b) these are mainly the Unorganized Territory and the unorganized parts of the Utah and New Mexico territories. The underlying shapefiles are taken from Manson et al. (2019) and the population data from Schroeder et al. (2026).

4.1 A Longitudinal Panel of County Populations

In this subsection, we use the area-based county crosswalk to create a longitudinal panel of population counts from 1790 to 2010 at the 2010 county level. The online repository associated with this paper (https://www.fpeckert.me/eglp) contains the code for one decade, 1900, in Python and Stata; the same steps apply to every decade.

We use data on historical population counts at the county level for every decade from 1790 to 2010 obtained from the NHGIS database (Schroeder et al., 2026). Once the data have been merged to the crosswalk, we aggregate population counts across the spatial subunits within each 2010 county for each decade from 1790 to 2010.

The procedure produces a longitudinal, unbalanced panel of US county populations between 1790 and 2010 at the 2010 county level. We calculate the total population that NHGIS assigns to counties in each decade and compare it to aggregate population counts for the US obtained from US Census publications. Since the crosswalk weights of each historical county sum to one, mapping these county counts to 2010 counties does not change their total. In most decades, the NHGIS county data and the Census totals agree to within 0.1%. In percentage terms, the largest discrepancy occurs in 1790, where the NHGIS county data account for 99.1% of the US population. Details are provided in Table E.1 in Appendix E.

Figure 6 shows the population data for 1850 on four maps. Panel (a) shows population density across all territories of the United States at the level of 1850 county delineations, and panel (b) shows the same population at the level of 2010 county delineations, mapped with our area-based crosswalk. Panels (c) and (d) do the same for the states that were part of the Union in 1850. The maps for 2010 counties show the same pattern as the 1850 maps, while spreading the population of large frontier counties over the smaller counties that later replaced them.

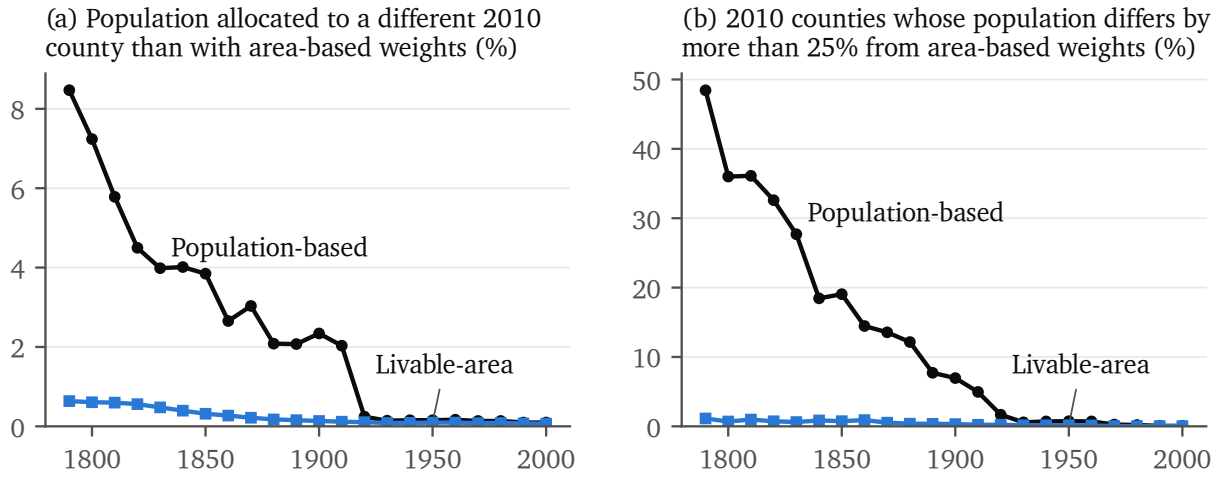
4.2 How Much Do the Weights Matter?

The three sets of weights rest on different assumptions about where people lived within a county. In this subsection, we compare each set of weights with data on where people lived that we do not use to construct the weights.

We measure the difference between two allocations of a historical county's population by the share of its population that they allocate to different 2010 counties. Let a_j and b_j denote the shares of the county's population that the two allocations assign to 2010 county j. The share of the population that they allocate to different 2010 counties is

$$D = \frac{1}{2} \sum_j \left| a_j - b_j \right| .$$

Figure 7: Differences between the Area-Based, Livable-Area, and Population-Based Weights



Notes: The figure compares the weights for 2010 county delineations in the contiguous United States in each decade. Panel (a) shows the share of the population in that decade's counties that the livable-area and population-based weights allocate to a different 2010 county than the area-based weights. Panel (b) shows the share of 2010 counties whose population in that decade under the livable-area or population-based weights differs by more than 25% from their population under the area-based weights. We count a 2010 county only if at least 100 people are allocated to it under the area-based weights. This rule excludes counties that overlap a historical county only in thin strips along county boundaries. The population data are taken from Schroeder et al. (2026).

Each person whom the two allocations place in different 2010 counties appears twice in the sum: once in the 2010 county to which the first allocation assigns more people and once in the 2010 county to which the second allocation assigns more people. The factor one half counts each such person once. We first use this measure to compare the three sets of weights with each other. We then use it to compare each set of weights with data on where people lived.

The choice of weights affects the allocation of the US population as a whole before 1920. Panel (a) of Figure 7 shows, for each decade, the share of the population that the livable-area and population-based weights allocate to a different 2010 county than the area-based weights. For the population-based weights, this share amounts to 8.5% in 1790 and 3.8% in 1850. It remains below 0.25% from 1920 onward. For the livable-area weights, this share amounts to 0.6% in 1790 and 0.3% in 1850. The shares are similar for 1990 county delineations (3.8% in 1850 for the population-based weights). They are smaller for 1990 commuting zones (0.9% in 1850), since these spatial units are larger.

The differences between the weights are larger for individual counties. Panel (b) of Figure 7 shows the share of 2010 counties whose population under a given set of weights differs by more than 25% from their population under the area-based weights. For the population-based weights, this share amounts to 48% in 1790, 19% in 1850, and 7% in 1900. For the livable-area weights, it is about 1% or less in all decades.

The differences between the three sets of weights are small from 1920 onward because few historical counties are split across 2010 counties after 1910. The weights matter only for historical counties whose territory is split across several 2010 counties. Table 2 shows that the fraction of crosswalk weights between 0.2 and 0.8 falls from 5.9% in 1850 to 1.8% in 1900 and 0.3% in 1920. From 1920 onward, nearly all the territory of a historical county lies in a single 2010 county, so all three sets of weights allocate its population in nearly the same way.

Since the three sets of weights differ before 1920, we compare each of them with data on where people lived within historical counties. For this comparison, we need a second measure of how the population of each historical county was split across 2010 counties. The censuses report only the total population of each county. The only consistent information on where people lived within a county that is available for *every* decade is the information our weights use: the location of towns and the rural population at later censuses. For some counties and decades, however, other sources show where people lived within the county. For these county-years, we compute the split directly from these sources and compare it with the split implied by each set of weights. We use three such sources, none of which is used to construct the weights.

Our first source of data on where people lived within counties is the Census Place Project (see Berkes et al., 2023), available on openICPSR (project 179401). The census records for each person the name of the place where the person lived, such as a town or a township. For each person in the full-count censuses from 1790 to 1940, the Census Place Project reports a person identifier and the latitude and longitude of the place that it matched to this name. From these coordinates, we determine two counties for each person: the county that contains them on the NHGIS county map of that census year, and the 2010 county that contains them. Counting people by pair of counties gives the population of each historical county in each 2010 county.[18]

Our second source of data on where people lived within counties combines the Census Place Project with the IPUMS USA full-count data for 1850 and 1880 (see Ruggles et al., 2025). The IPUMS data contain the digitized census record of every person. For each person, they report the state and county in which the census enumerated the person (the variables STATEICP and COUNTYICP) and the same person identifier as the Census Place Project (HISTID). We use these data because the coordinates of the Census Place Project do not always lie in the county in which the person was enumerated. The Census Place Project matches place names to modern lists of places, and some matches are imprecise. For 2.0% of the people in 1850 and 1.4% in 1880, the coordinates lie in a different county than the IPUMS county. About 40% of these people lie within five kilometers of their IPUMS county, at places near a county boundary; most of the others lie more than 20 kilometers away. We link the two sources by the person identifier. Each person's

[18]The Census Place Project reports only household heads before 1850 and has no data for 1890. We do not use it to construct the weights because it does not report the distribution of people within counties. It locates all people recorded in a place, including rural residents, at a single point.

Table 4: Population Allocated to a Different 2010 County than in the Comparison Data (%)

Comparison data	Area	Livable Area	Population
Census Place Project, 1790–1940	5.04	4.92	2.66
Census Place Project with IPUMS counties, 1850	6.56	6.44	3.92
Census Place Project with IPUMS counties, 1880	3.73	3.67	2.17
Township counts, 1790–1860	8.35	8.25	3.63

Notes: The table shows the share of the population that each set of weights allocates to a different 2010 county than the comparison data. Within a census year, each county is weighted by its census count. The first row computes this share separately for each of the 15 census years from 1790 to 1940 (1890 is missing) and reports the simple mean of the 15 shares. In each year, it includes every county in which the Census Place Project locates at least 500 people who account for at least half of its census count (a tenth before 1850). The second and third rows each use a single census year, 1850 and 1880, with the same rule for including counties. The last row pools all 105 county-years from 1790 to 1860 in which a county of New Hampshire, Vermont, New York, or Maine covers parts of more than one 2010 county. Each of these county-years is weighted by its census count. Appendix F describes the data sources and samples.

historical county then comes from the IPUMS data, and the 2010 county still comes from the coordinates. These data cover 840 counties in 1850 and 1,939 in 1880.

Our third source of data on where people lived within counties, the census counts of townships in four northeastern states, does not rely on coordinates. In New Hampshire, Vermont, New York, and Maine, the land is divided into townships, which these states call towns. Townships cover the entire territory, and the census reports the population of each. We take the township populations from the historical census tables of New Hampshire Office of Strategic Initiatives (2026) for New Hampshire and Vermont Center for Geographic Information (2026) for Vermont (1790–1860), from Hough (1857) for New York (1790–1850), and from De Bow (1853) for Maine (1850). We match each township to its county subdivision in the 2010 TIGER/Line files of the U.S. Census Bureau (2010). The county subdivision gives the township's 2010 county, and the county that contains the township on the NHGIS county map of that census year gives its historical county. Summing the township populations by pair of counties gives the split of each county that was later divided. These data cover 105 county-years in 37 counties with 3.1 million people.

For each set of weights, Table 4 reports the share of the population that the weights allocate to a different 2010 county than the comparison data, the measure D above. Each county is weighted by its census population count, and all three sets of weights are evaluated on the same counties. The first row uses the Census Place Project. The second and third rows use the Census Place Project with the IPUMS counties, and the last row uses the township counts.

The population-based weights differ least from the comparison data in every row of Table 4. The

share of the population that they allocate to a different 2010 county than the comparison data is between 43% and 60% of the share under the area-based weights. The livable-area weights lower the share relative to the area-based weights by about one tenth of a percentage point.

Taking the historical county of each person from the IPUMS data raises the shares in 1850 and 1880 by about 0.8 percentage points for each set of weights, compared with the Census Place Project alone. The order of the three sets of weights does not change.

The difference between the population-based and the area-based weights has two sources: the location of towns and the distribution of the rural population. To separate the two, we construct one more set of weights for population. These weights place the towns as the population-based weights do. They spread the rural population of each county evenly over its livable land in every decade, as in step 2 of Figure 5. Table 4 does not include these weights. We report their results only in the text.

In the comparison with the Census Place Project, the location of towns and the distribution of the rural population account for similar parts of the difference between the area-based and the population-based weights. The first row of Table 4 uses these data. It reports that the area-based weights allocate 5.04% of the population to a different 2010 county than the comparison data, and the population-based weights 2.66%. The population-based weights thus lower the share by 2.38 percentage points (5.04 minus 2.66). The weights that place the towns but spread the rural population evenly allocate 3.95%. Placing the towns therefore lowers the share from 5.04% to 3.95%, by 1.09 percentage points. Spreading the rural population in proportion to the rural population at later censuses, as the population-based weights do, lowers it further from 3.95% to 2.66%, by 1.29 percentage points.

In the comparison with the township counts, the distribution of the rural population accounts for most of the difference between the area-based and the population-based weights. The last row of Table 4 uses these data. It reports that the area-based weights allocate 8.35% of the population to a different 2010 county than the comparison data, and the population-based weights 3.63%. The population-based weights thus lower the share by 4.72 percentage points (8.35 minus 3.63). The weights that place the towns but spread the rural population evenly allocate 7.48%. Placing the towns therefore lowers the share from 8.35% to 7.48%, by 0.87 percentage points. Spreading the rural population in proportion to the rural population at later censuses lowers it further from 7.48% to 3.63%, by 3.85 percentage points.

5. Concluding Remarks

In this paper, we propose and implement a simple method to construct crosswalks between a reporting and a reference map. The method relies on intersecting two maps to create a new map

consisting of smaller spatial subunits. For each subunit, we compute the share it accounts for in the overall area of the spatial unit that nests it in the reporting map. The collection of these shares, or "weights," constitutes a crosswalk from the reporting to the reference map. The weights can be used to re-aggregate data on stocks (e.g., population counts) available for the reporting units to the reference units.

We apply the method to US counties since 1790. We construct a simple-to-use crosswalk that researchers can employ to map historical county-level data for the United States to present-day county delineations or commuting zones. We then apply the crosswalk to historical census data to create a longitudinal panel of county populations on the 2010 county map. Lastly, we extend our approach by computing the same shares with two other measures. The first is livable area, which uses information on natural terrain. The second is estimated population, which uses information on the size and location of towns.

A data repository associated with this project at https://www.fpeckert.me/eglp contains the following material: (i) Python code that allows researchers to take two maps of the same territory and construct their own crosswalk (see Appendix B for step-by-step instructions), (ii) Python code and crosswalks that map historical US counties since 1790 to 1990 counties, 2010 counties, or 1990 commuting zones, with area-based weights and, for the contiguous United States, with livable-area and population-based weights, and (iii) as an example of how to use the county crosswalk, Python and Stata code that maps 1900 county-level population counts to 2010 county delineations.

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Appendix

A. Details of County Crosswalk Construction

In this appendix, we discuss some details of the crosswalk construction.

FIPS, NHGIS, and ICPSR County Codes Historical data downloaded from NHGIS or other sources often contain several sets of county and state identifiers. The three main classification schemes are FIPS, NHGIS, and ICPSR codes. The FIPS codes are the standard codes used in most modern data sets. NHGIS codes extend them to historical units: for counties that have a FIPS code, the NHGIS code is the FIPS code with a zero appended, and counties, territories, and other areas that no longer exist receive NHGIS codes of their own. ICPSR codes are a coding scheme developed by the Inter-university Consortium for Political and Social Research (ICPSR) and are used in the ICPSR historical county data. ICPSR *state* codes always differ from FIPS state codes. ICPSR *county* codes usually equal the NHGIS county code, i.e., ten times the FIPS county code. This holds for 3,101 of the 3,128 counties of 1990 that have an ICPSR code; the exceptions are mostly in Maryland, where ICPSR keeps an older numbering. Not every unit has an ICPSR code. In our crosswalks, 13 of the 3,141 counties of 1990 have no ICPSR code. Several of them were created after 1980. In the other decades from 1790 to 2000, 564 records of counties and other units have no ICPSR code, most of them in territories or in areas the census did not enumerate. Every unit has an NHGIS code. We therefore recommend merging data to the crosswalk on NHGIS codes, or on FIPS codes for counties that have them, and using ICPSR codes only for data that come with ICPSR codes alone.

Shapefile Imprecisions We use the Python GeoPandas package to overlay shapefiles for different decades and form their intersection. However, some of the shapefiles contain imprecisions. Sometimes, even county boundaries that have not changed between two decades do not exactly overlap. In other words, the geographical delineations of some unchanged counties differ slightly between two generations of shapefiles. In such cases, our procedure identifies very small weights that would be added as additional entries in the crosswalk. In practice, it is difficult to determine which intersections identify actual county boundary changes and which are just due to such imprecisions. To be conservative, we drop intersections that create areas of ten square meters or less in the construction of our crosswalk. The crosswalk *weights* associated with these dropped areas are typically of the order of 10^{-8} and never exceed 10^{-4}, i.e., they account for a very small share of the areas of reporting counties. The map of 2010 counties is drawn from different line work (the 2010 TIGER/Line shapefiles) than the maps of earlier decades (the 2000 TIGER/Line shapefiles). Even counties that did not change between 2000 and 2010 therefore intersect

neighboring 2010 counties in thin slivers that are larger than ten square meters. These slivers remain in the 2010 county crosswalk. They carry very small weights (for the counties of 2000, a median of 0.03% of each county's area lies outside its main 2010 county) and do not affect mapped totals. The commuting zone map is drawn from yet other line work, so the commuting zone crosswalk contains similar slivers. Where two maps draw a coastline differently, a small part of a historical county can also lie outside the reference map; the weights of the part inside are then scaled up to sum to one.

Land and Water The NHGIS county maps follow the shoreline: they exclude the ocean, bays, and the Great Lakes. They do include the inland lakes, reservoirs, and wide rivers that lie within county boundaries, which make up about 1.8% of the area of the 2010 map. The weights are therefore shares of land and inland water. Removing the inland water would change little. By a rough calculation that spreads each 2010 county's inland water evenly over the county, it would allocate about 0.03% of the 1900 US population to a different 2010 county and change the 1900 population of 5 of the 3,143 counties of 2010 by more than 10%. For historical data, removing it would also be questionable, because many of the largest inland water bodies are reservoirs built in the twentieth century on land that was inhabited before.

Counties, Parishes, and Independent Cities In earlier years, counties did not cover the entire territory of the US. Instead, some parts were covered by parishes, independent cities, and other subdivisions. In fact, some independent cities continue to exist to this day.[19] The crosswalk maps these subdivisions to 2010 counties or independent cities. NHGIS assigns all these subdivisions "county" codes of their own, and most of them also have ICPSR codes. Only counties and independent cities that still exist in 2010 have "county" identifiers in 2010. Researchers who map data for such non-county subdivisions should therefore merge on the NHGIS codes of the historical units.

[19] Formally, an independent city is not part of the territory of any county. Today there are 41 independent US cities, 38 of which are in Virginia; the 2010 map has 42, because Bedford City, Virginia, was still independent. The three independent cities outside Virginia are Baltimore (Maryland), St. Louis (Missouri), and Carson City (Nevada). The US Census Bureau uses counties as its base unit for presentation of statistical information, and treats independent cities as county equivalents for those purposes.

B. How to Create Customized Crosswalks

Here we describe how to use our programs to create geographical crosswalks between a reporting and a reference map of the same territory.

- Go to https://www.fpeckert.me/eglp and download the Python code for creating general geographical crosswalks.
- Ensure that the reporting and reference shapefiles each have a coordinate system (a `.prj` file; the code projects both maps to a common equal-area projection) and cover the same territory (e.g., the contiguous United States) but contain different partitions of it (e.g., counties and school districts). Note that each shapefile must contain a column that is the unique identifier of each region, e.g., state and county codes in the case of our county crosswalk.
- Copy the *file paths* to the reporting and reference shapefiles into the indicated locations in the Python file (i.e., fill in values for the placeholders "PATH_REPORTING", "PATH_REFERENCE").
- Copy the *file names* of the shapefiles into the indicated locations in the Python file (i.e., fill in values for the placeholders "REPORTING_FNAME", "REFERENCE_FNAME").
- Copy the *column names* of the unique identifier in the two shapefiles into the indicated locations in the Python file (i.e., fill in values for the placeholders "REPORTING_GEOID" and "REFERENCE_GEOID").
- Fill in the folder and file name for the output (placeholders "OUTPUT_PATH", "OUTPUT_FNAME").
- Run the code. For detailed shapefiles (e.g., counties instead of states) the procedure is time-consuming.
- The output produced by the program is a comma-separated values (CSV) file with three columns: (1) the reporting unique identifier, (2) the reference unique identifier, and (3) the associated weights.

C. Description of Crosswalk Files

The paper's website, https://www.fpeckert.me/eglp, links to the repository https://github.com/liang-jack-a/EGLP_Crosswalk, which holds three versions of the crosswalk. They map county delineations from 1790 to 2010 to (1) 1990 county delineations, (2) 2010 county delineations, and (3) 1990 commuting zone delineations (Tolbert and Sizer, 1996). Each crosswalk comes as a comma-separated values (CSV) file and as a zipped Stata file. It contains the geographical identifiers of the historical units for each decade, the identifiers of the reference units, and a set of weights. Tables C.1, C.2, and C.3 list the variables in each crosswalk.

Using the crosswalk takes four steps.

1. Choose the reference map (1990 counties, 2010 counties, or 1990 commuting zones) and keep the rows of the crosswalk for the census year closest to the data.
2. Merge the data onto the crosswalk. Data from NHGIS carry the identifier `GISJOIN`, which the crosswalk contains as well. Other data can be merged on the NHGIS state and county codes: for counties that have a FIPS code, the NHGIS code is the FIPS code with a zero appended (e.g., FIPS state code 51 is NHGIS state code 510, and FIPS county code 650 is NHGIS county code 6500). Each historical county matches one row for each reference unit it overlaps.
3. Multiply the variable of interest, e.g., the population count, by the weights.
4. Sum the result within the reference units, identified by their five-digit FIPS codes (`FIPS_1990`, `FIPS_2010`) or by their commuting zone codes (`cz`).

In Stata, with 1900 county population from NHGIS saved in `mydata.dta` (one row per county, with `GISJOIN` and `population`) and the crosswalk unzipped with `unzipfile`, the four steps read:

```
use if Year == 1900 using "county_crosswalk_endyr_2010.dta", clear
merge m:1 GISJOIN using "mydata.dta", keep(match) nogenerate
generate double population_2010 = population * weight
collapse (sum) population_2010, by(FIPS_2010)
```

Doing this for each decade produces a longitudinal panel at the reference delineations. The repository contains this example as complete scripts in Python and Stata, using historical population data from Manson et al. (2019).

The crosswalk can also be used to map data on average outcomes rather than stocks (e.g., average wages rather than total payroll). To this end, we include the "area" variable in each crosswalk. It contains the area of the overlap between the reporting and the reference county. For data on

averages, the correct weight is the share of the area of the reference county accounted for by the reporting county. Having computed these weights, researchers can re-weight averages for historical counties to obtain them at the level of the reference counties.

Each crosswalk also contains the livable-area and population-based weights of Section 3.3, in the variables `weight_livable` and `weight_population`. They cover the contiguous United States and are empty for Alaska and Hawaii. To use them, researchers multiply by `weight_livable` or `weight_population` instead of `weight` in the third step above.

Table C.1: Variables in the 1990 County Crosswalk

Variable	Description
Year	Decade of the historical data; takes values 1790–2010 except 1990
NHGISST	NHGIS state code of historical county
NHGISCTY	NHGIS county code of historical county (three-digit FIPS code in the rows for 2010)
GISJOIN	NHGIS identifier of historical county, as in NHGIS data extracts (e.g., G0100010)
STATENAM	Name of state of historical county
NHGISNAM	Name of historical county
ICPSRST	ICPSR state code of historical county
ICPSRCTY	ICPSR county code of historical county
area_base	Area, in square meters, of the historical county
NHGISST_1990	NHGIS state code of reference county
NHGISCTY_1990	NHGIS county code of reference county
STATENAM_1990	Name of state of reference county
NHGISNAM_1990	Name of reference county
ICPSRST_1990	ICPSR state code of reference county
ICPSRCTY_1990	ICPSR county code of reference county
GISJOIN_1990	NHGIS identifier of reference county
FIPS_1990	Five-digit FIPS code of reference county (e.g., 01001)
area	Area, in square meters, of the intersection between the historical county and the reference county
weight	Share of historical county's area that overlaps with reference county in 1990, among the parts that lie on the reference map (the weights of each historical county sum to one)
weight_livable	Share of historical county's livable land in the reference county (Section 3.3; contiguous United States only)
weight_population	Share of historical county's population in the reference county (Section 3.3; contiguous United States only)
US_STATE	Indicator for whether the US state containing the historical county was part of the Union in the year (always 0 for the District of Columbia)

Notes: The table shows the variables in the crosswalk that maps US counties from 1790 to 2010 to 1990 county delineations (crosswalk available at https://www.fpeckert.me/eglp).

Table C.2: Variables in the 2010 County Crosswalk

Variable	Description
Year	Decade of the historical data; takes values 1790–2000
NHGISST	NHGIS state code of historical county
NHGISCTY	NHGIS county code of historical county
GISJOIN	NHGIS identifier of historical county, as in NHGIS data extracts (e.g., G0100010)
STATENAM	Name of state of historical county
NHGISNAM	Name of historical county
ICPSRST	ICPSR state code of historical county
ICPSRCTY	ICPSR county code of historical county
area_base	Area, in square meters, of the historical county
NHGISST_2010	NHGIS state code of reference county
NHGISCTY_2010	FIPS county code (three digits) of reference county
STATENAM_2010	Name of state of reference county
NHGISNAM_2010	Name of reference county
ICPSRST_2010	Empty; ICPSR codes are not available for 2010 counties
ICPSRCTY_2010	Empty; ICPSR codes are not available for 2010 counties
GISJOIN_2010	NHGIS identifier of reference county
FIPS_2010	Five-digit FIPS code of reference county (e.g., 01001)
area	Area, in square meters, of the intersection between the historical county and the reference county
weight	Share of historical county's area that overlaps with reference county in 2010, among the parts that lie on the reference map (the weights of each historical county sum to one)
weight_livable	Share of historical county's livable land in the reference county (Section 3.3; contiguous United States only)
weight_population	Share of historical county's population in the reference county (Section 3.3; contiguous United States only)
US_STATE	Indicator for whether the US state containing the historical county was part of the Union in the year (always 0 for the District of Columbia)

Notes: The table shows the variables in the crosswalk that maps US counties from 1790 to 2010 to 2010 county delineations (crosswalk available at https://www.fpeckert.me/eglp).

Table C.3: Variables in the 1990 Commuting Zone Crosswalk

Variable	Description
Year	Decade of the historical data; takes values 1790–2010
NHGISST	NHGIS state code of historical county
NHGISCTY	NHGIS county code of historical county (three-digit FIPS code in the rows for 2010)
GISJOIN	NHGIS identifier of historical county, as in NHGIS data extracts (e.g., G0100010)
STATENAM	Name of state of historical county
NHGISNAM	Name of historical county
ICPSRST	ICPSR state code of historical county
ICPSRCTY	ICPSR county code of historical county
area_base	Area, in square meters, of the historical county
cz	Code of the reference commuting zone of 1990 (see Tolbert and Sizer, 1996)
area	Area, in square meters, of the intersection between the historical county and the reference commuting zone
weight	Share of historical county's area that overlaps with reference commuting zone in 1990, among the parts that lie on the reference map (the weights of each historical county sum to one)
weight_livable	Share of historical county's livable land in the reference commuting zone (Section 3.3; contiguous United States only)
weight_population	Share of historical county's population in the reference commuting zone (Section 3.3; contiguous United States only)
US_STATE	Indicator for whether the US state containing the historical county was part of the Union in the year (always 0 for the District of Columbia)

Notes: The table shows the variables in the crosswalk that maps US counties from 1790 to 2010 to 1990 commuting zone delineations as defined by Tolbert and Sizer (1996) (crosswalk available at https://www.fpeckert.me/eglp).

D. An Alternative County Crosswalk

Horan and Hargis (1995) developed the County Longitudinal Template (CLT) for the years from 1840 to 1990. The CLT traces the historical origins of counties to create county groups that can be studied over time. For each decade, it assigns every county of 1990 to a cluster: the counties whose data must be added up so that the cluster covers the same territory in that decade and in every later one. In this case no weights are needed; count variables are summed across the counties of a cluster in every decade.

Table D.1 and Figure D.1 compare the two approaches for Montgomery County, Alabama, in 1840. In the CLT, Montgomery belongs to a cluster of 29 counties of 1990 (Panel B of the table; gray and blue in the figure). To compare 1840 with any later decade, a researcher must add up the data of all 29 counties, which together cover 20 counties of 1840. Repeated splits and boundary changes after 1840 joined these counties into one cluster, which covers far more than the area of historical Montgomery: it includes, for example, Autauga, Barbour, and Butler counties, which share no area with it.

Our crosswalk assigns the data of 1840 Montgomery to three counties of 1990 (Panel A of the table; blue in the figure). Modern Montgomery County receives about 0.80, reflecting the fact that historical Montgomery County was situated approximately where modern Montgomery County is; Elmore receives about 0.12 and Bullock about 0.08. The other counties of 1840 in the cluster

Table D.1: Montgomery County (AL) in 1840 in Two Crosswalks

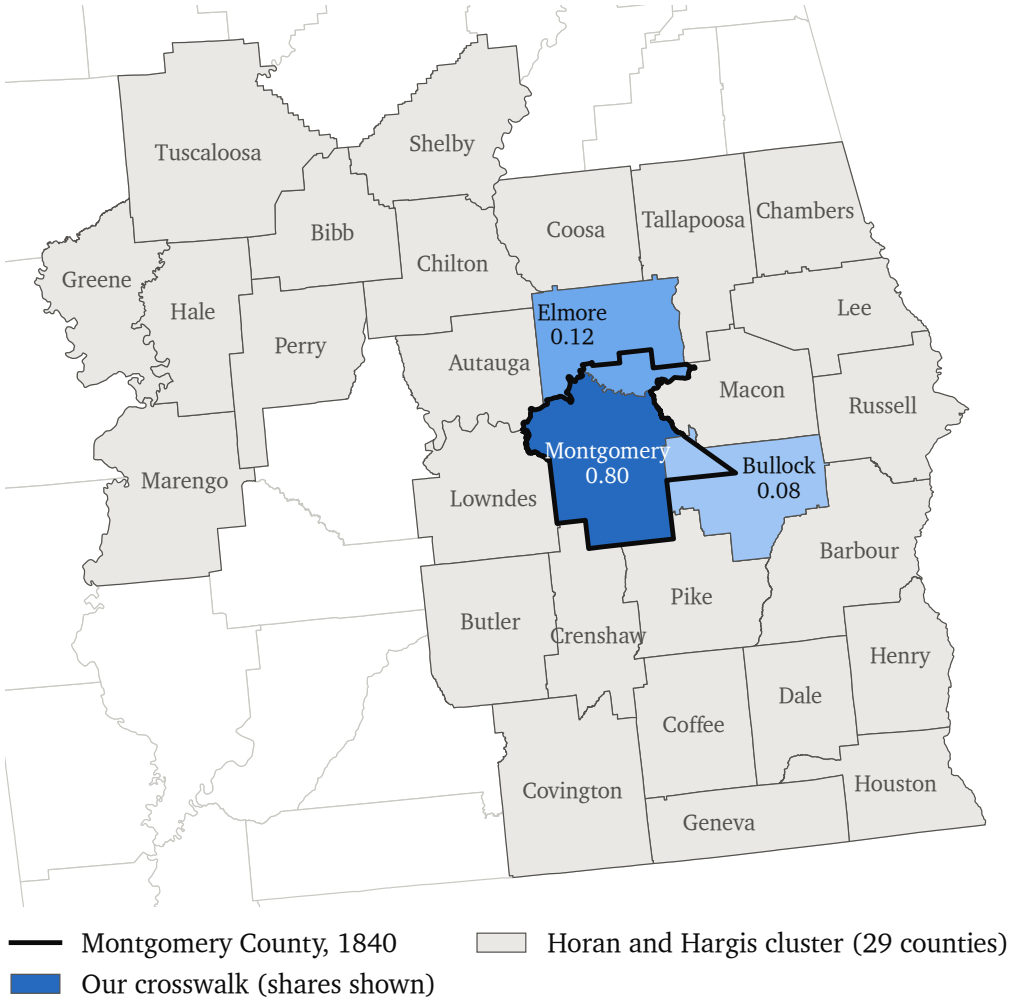
Panel A: Our crosswalk	
1990 County	Share of 1840 Montgomery
Montgomery	0.80
Elmore	0.12
Bullock	0.08
Panel B: County Longitudinal Template	
The 29 counties of 1990 in the cluster that contains Montgomery in 1840, which must be added up for every decade: Autauga, Barbour, Bibb, Bullock, Butler, Chambers, Chilton, Coffee, Coosa, Covington, Crenshaw, Dale, Elmore, Geneva, Greene, Hale, Henry, Houston, Lee, Lowndes, Macon, Marengo, Montgomery, Perry, Pike, Russell, Shelby, Tallapoosa, Tuscaloosa. Together they cover 20 counties of 1840.	

Notes: Panel A shows how our area-based crosswalk distributes the data of Montgomery County in 1840 across the counties of 1990: the shares of its area that lie in each of them. Panel B shows the cluster of the County Longitudinal Template of Horan and Hargis (1995) that contains Montgomery County in 1840. The underlying shapefiles are taken from Manson et al. (2019).

are mapped in the same way, so an analysis can stay at the level of the counties of 1990 instead of a single cluster of 29 of them.

Our crosswalk has a range of additional advantages. We extend the crosswalk back 50 years to 1790 and forward 20 years to 2010. Our crosswalk can be used to work with present-day counties and does not necessitate working with historical county groups. With the CLT, the further back in time researchers go, the more data for *all* years have to be aggregated, since data always have to be summed to the larger historical clusters. With our crosswalk, all data for all decades can be mapped to present-day county boundaries. Furthermore, because our crosswalk reports the area of every subunit, it can be used in both directions: to disaggregate data for historical counties to modern county or commuting zone delineations, and, under the same assumption of proportionality to area, to aggregate data reported for present-day counties to the delineations of earlier years.

FIGURE D.1: MONTGOMERY COUNTY (AL) IN 1840 IN TWO CROSSWALKS



Notes: The thick black line shows Montgomery County in 1840. The blue counties of 1990 are those to which our crosswalk assigns the data of 1840 Montgomery, with their shares. The gray counties of 1990, together with the blue ones, form the cluster that contains Montgomery in 1840 in the County Longitudinal Template of Horan and Hargis (1995). The underlying shapefiles are taken from Manson et al. (2019).

E. Additional Tables

Table E.1: The Aggregate Population in the Census and from NHGIS

			Discrepancy	
Year	Census	NHGIS	Absolute	Percentage (%)
1790	3,929,214	3,893,934	35,280	0.9
1800	5,308,483	5,299,313	9,170	0.2
1810	7,239,881	7,238,667	1,214	0.0
1820	9,638,453	9,621,616	16,837	0.2
1830	12,860,702	12,860,492	210	0.0
1840	17,063,353	17,063,353	0	0.0
1850	23,191,876	23,191,876	0	0.0
1860	31,443,321	31,362,622	80,699	0.3
1870	38,558,371	38,557,210	1,161	0.0
1880	50,189,209	50,011,451	177,758	0.4
1890	62,979,766	62,654,302	325,464	0.5
1900	76,212,168	75,902,898	309,270	0.4
1910	92,228,496	92,164,175	64,321	0.1
1920	106,021,537	106,021,568	−31	0.0
1930	123,202,624	123,202,660	−36	0.0
1940	132,164,569	132,165,129	−560	0.0
1950	151,325,798	151,325,798	0	0.0
1960	179,323,175	179,323,175	0	0.0
1970	203,211,926	203,211,926	0	0.0
1980	226,545,805	226,545,805	0	0.0
1990	248,709,873	248,709,873	0	0.0
2000	281,421,906	281,421,906	0	0.0
2010	308,745,538	308,745,538	0	0.0

Notes: The table reports total US population in each decade and the total population that NHGIS assigns to the counties and other units on that decade's map. Columns 4 and 5 also show the discrepancy between aggregate population numbers and what can be attributed to a county using the NHGIS data. The NHGIS data are taken from Schroeder et al. (2026). The Census data for 1790–1990 are taken from Forstall (1996); those for 2000 and 2010 are the official census counts. Following Forstall (1996), the Census totals cover the territory of today's 50 states and the District of Columbia, including Alaska from 1880 on and Hawaii from 1900 on, and exclude persons abroad. For 1970 they are the counts before the later corrections, which are also the counts in the NHGIS data. The NHGIS county population data were downloaded in November 2026.

Table E.2: Accession of US States to the Union

State	Union Accession	State	Union Accession
Delaware	1787	Michigan	1837
Pennsylvania	1787	Florida	1845
New Jersey	1787	Texas	1845
Georgia	1788	Iowa	1846
Connecticut	1788	Wisconsin	1848
Massachusetts	1788	California	1850
Maryland	1788	Minnesota	1858
South Carolina	1788	Oregon	1859
New Hampshire	1788	Kansas	1861
Virginia	1788	West Virginia	1863
New York	1788	Nevada	1864
North Carolina	1789	Nebraska	1867
Rhode Island	1790	Colorado	1876
Vermont	1791	North Dakota	1889
Kentucky	1792	South Dakota	1889
Tennessee	1796	Montana	1889
Ohio	1803	Washington	1889
Louisiana	1812	Idaho	1890
Indiana	1816	Wyoming	1890
Mississippi	1817	Utah	1896
Illinois	1818	Oklahoma	1907
Alabama	1819	New Mexico	1912
Maine	1820	Arizona	1912
Missouri	1821	Alaska	1959
Arkansas	1836	Hawaii	1959

Notes: The table reports the year in which individual states became part of the Union of the United States, i.e., became US states as we know them today. States are ordered by their date of accession to the Union. In the crosswalk, the indicator US_STATE equals 1 for a state from the first decennial census in or after its year of accession. California (admitted September 9, 1850), Idaho (July 3, 1890), and Wyoming (July 10, 1890) joined the Union one to about three months after census day, but California is counted as a state in 1850, and Idaho and Wyoming in 1890.

F. Details of the Livable-Area and Population-Based Crosswalks

In this appendix, we discuss some details of the crosswalks in Section 3.3 and the comparisons in Section 4.2. The folder `towns` in our repository contains the code to construct the crosswalks and comparisons from the raw data. Its documentation lists all data sources.

Livable Land We obtain elevation and slope from the 7.5 arc-second mean elevation data in Danielson and Gesch (2011). We obtain water bodies from the National Atlas (see National Atlas of the United States, 2014), which include lakes, intermittent and dry lakes, glaciers, swamps, and marshes. We classify a lake as a reservoir if a major dam at least 40 feet high lies within three kilometers and the listed reservoir area of the dam is at least a quarter of the area of the lake (see National Atlas of the United States, 2006). We treat reservoirs as land. We classify land on a finer grid of 250-meter squares and compute the livable share of each grid cell of Section 3.3.

Towns We take town populations in each decade from Steiner and Heppler (2015), who compile them mainly from a Census Bureau list of places that passed 2,500 people. Smaller places therefore count as rural population in our construction. We take town locations from U.S. Geological Survey (2026), whose coordinates mark traditional town centers. We place each town in the nearest grid cell, within ten kilometers of its coordinates, that lies in the town's state and in the same historical county and 2010 county as its coordinates. Some cities form their own unit on the census map, such as the independent cities of Virginia. We place the population of such a city inside its unit. We scale down a town whose population exceeds the census count of its county to that count.

Rural Population and Weights Several rules apply to the rural population and to the final weights. Grid cells in a county that had at least 99% of its population in towns in the following census carry no information on the distribution of rural population. They receive the average rural population per unit of livable land of the other grid cells of their county in the current decade. We set weights below one in a million to zero and rescale the remaining weights. In the comparisons of Section 4.2, we measure the area-based weights on the same grid cells, counting each grid cell in the county in which its center lies. They therefore allocate up to 0.06% of the population differently than the area-based crosswalks in Section 3, which intersect the maps exactly.

Thin Strips along County Boundaries The livable-area and population-based weights give no weight to thin strips along county boundaries. The historical maps and the reference map do not trace the same county boundaries exactly, even where a boundary did not change (see Appendix A). Their intersection therefore contains, for almost every county, thin strips along its boundary that lie in a neighboring county of the reference map. The area-based weights give each strip a

weight in proportion to its area, which is tiny. On the grid, however, a strip only a few meters wide can contain the centers of several grid cells and therefore receive their land and their people. The livable-area and population-based weights would then allocate these people to the neighboring county. To avoid this, we identify the strips by their shape: a subunit counts as a strip if all of its grid cells lie within one kilometer of a boundary between two counties on either map. Coastlines do not count as boundaries, so islands are not strips. A strip receives a weight of zero, and the other weights of its county are scaled up so that they still sum to one. We keep a strip if it contains a town, since the town places people in it. In these comparisons, the area-based weights keep the strips.

Corrections to the Data The NHGIS county maps and the census county counts sometimes disagree. We correct the population-based weights where the census record is clear. For example, the census counted suburbs of New Orleans in Jefferson Parish from 1840 to 1860. The maps draw Orleans Parish at its 2010 extent. We therefore assign the suburbs' share of Jefferson Parish to Orleans Parish. The other corrections concern duplicate or misplaced towns and cities counted within a neighboring county (e.g., Petersburg, Virginia, and Trenton, New Jersey). The code documents each correction. The documentation in the repository lists all corrections and the errors we know remain. The largest remaining error is that land later annexed by a city receives too little population before the annexation. In the following census, the people on this land count toward the city, which we place in a single grid cell, so the land itself shows little rural population. The shortfall is typically 5,000 to 25,000 people in a given county and decade. The crosswalks do not cover units for which the census reports a population but which do not appear on the map of that decade. These units amount to about 19,300 people in the 1900 Indian reservations and a few short-lived frontier counties.

Data Sources for the Comparisons The three data sources in Section 4.2 differ in their samples and coverage. In the comparisons based on Berkes et al. (2023), we include a county if the data locate at least 500 of its people and they account for at least half of its census count, or a tenth before 1850. Before 1870, these data contain only free people. Before 1850, they contain only household heads. The returns of the 1890 census were almost entirely destroyed. These comparisons therefore cover 15 decades. The comparisons with the IPUMS counties use the full-count data for 1850 and 1880 in Ruggles et al. (2025). They exclude Dakota Territory in 1880, about 0.3% of the population. We transcribed the New York and Maine township tables from scans of the printed reports. The transcriptions are available in the repository. The townships of each county add up to its printed county total.